

Machine learning and public policy: Early detection of physical violence against children[☆]

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ABSTRACT

Physical violence against children is a widespread and grossly underreported phenomenon with substantial short and long-term negative consequences. In Latin America and the Caribbean, 43% of children under the age of 15 experience corporal punishment at home, yet reporting rates are alarmingly low. This paper aims to demonstrate how household data can be considered for a future predictive analytics model in Argentina. Based on the 2019–20 MICS survey we apply machine learning techniques to predict physical violence against children (understood as physical discipline) at the household level in Argentina. The scope and potential benefits of using predictive models in this context are assessed, as well as the main limitations and risks. The results suggest that, by analyzing the situation of the 30% of households with the highest risk scores, 43 out of 100 households in which children experience physical violence could be identified at an early stage.

1. Introduction

The use of violence as a method of discipline is a widespread phenomenon. In Argentina, according to the Multiple Indicator Cluster Survey (MICS) conducted by the United Nations Children's Fund (UNICEF) in 2019–2020, 59.4 % of children under the age of 15 experienced some form of disciplinary violence (psychological or physical) at home in the month prior to the interview. During the same period, 35.4 % were subjected to corporal punishment (whether severe or not) and 6.6 % were subjected to severe physical punishment.¹ Although the prevalence of violent discipline in the home has decreased compared to the previous MICS survey (2011–2012), it remains high.

Despite being widespread, reporting levels of physical violence on children are alarmingly low. According to Stoltenborgh et al. (2013), only 1 in 75 cases of child physical abuse and 1 in 30 cases of sexual abuse are reported worldwide.

To assess a child's risk of experiencing violence and provide input into the decision-making processes, the two main approaches

traditionally used in child protection settings are theoretical or consensus-based models, and actuarial models. Both approaches take into account a set of family characteristics that are associated with child maltreatment, but the main differences between these models lie in the process by which the factors are selected and how the models are used in practice (D'Andrade et al., 2008). A third approach, more recently adopted in the context of child protection, is the use of predictive analytics. This methodology aims to discover patterns and make predictions by applying modeling and analytical techniques to existing data (Cuccaro-Alamin et al., 2017). Applications of these techniques can be found in the insurance, credit, and healthcare sectors. Over the past decade, several initiatives and projects have emerged that use predictive analytics in child protection settings, with the goal of contributing to the early detection of violence and increasing and improving the efficiency of resource allocation.

This study contributes to the debate on the use of predictive analytics in child protection. We explore the application of machine learning techniques to predict the risk of physical violence against children

[☆] This work stemmed from the Caring Project (the real name is not included for confidentiality reasons), an initiative that incorporates predictive analysis in the implementation of public policies in Argentina. This work has aimed to improve and strengthen the predictive models to increase their prediction accuracy.

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¹ Physical punishment is considered to be: 1) "spanked, hit, slapped on the bottom with bare hand", 2) "hit with a belt, hairbrush, stick or another hard object", 3) "hit/slapped on hand, arm or leg", 4) "hit/slapped on the face, head or ears", and 5) "beat up, hit over and over again as hard as you can". The last two statements were considered severe physical punishment.

(understood as corporal punishment) at the household level in Argentina. Our aim is to contribute to public policy design by demonstrating the feasibility of developing a model capable of conducting large-scale initial screenings to identify instances of child violence at early stages. This involves incorporating an additional preliminary step into the standard process that typically starts after a case has been reported. These predictive models could expand the scope of early detection by identifying children who may be at risk, thereby mitigating issues related to underreporting and bias. However, it's important to clarify that our objective is not to offer insights or directives on how individual cases should be handled. Rather, this model is not intended to supplant the regular process through which social workers address each case, but rather to broaden the pool of children identified as potential victims. Ultimately, the social worker overseeing each case will be able to discern between correct and incorrect predictions made by the model.²

The effectiveness of the machine learning models hinges on access to extensive information about the child and their environment. While such data is likely available within Argentina's child protection system administrative records, it is not publicly accessible. Therefore, we developed the models using the MICS survey, a nationally representative survey conducted by UNICEF in Argentina in 2019–20. In this way, we aim to show how rich data may be used to extend the reach of early detection efforts. Additionally, our work underscores the importance of integrating information from various public agencies, even if it is not immediately suitable for this purpose. Furthermore, it may reveal gaps in existing data collection practices and highlight areas where additional data collection efforts are needed.

Our results show that the models are able to identify 43 % of households in which children experience physical violence, within the 30 % of households with the highest risk. Although this result represents an improvement over the current *status quo* and could be used by social workers to broaden the pool of potential children at risk, further work is needed to reach the remaining 57 % of households where physical violence is not correctly identified.

Three important matters should be noted. First of all, predicting physical violence against children is highly complex and the methodologies presented in this study are not able to overcome all the difficulties in place. Therefore, the scope and potential benefits, as well as the main limitations and risks, of the proposed methodologies are discussed throughout the paper. Secondly, the development of the models was guided by considerations related to the actual implementation of the instrument in the field. This implied that ethical and even operational issues drove many of the decisions undertaken, for instance, the use of strategies to mitigate stigmatization and leaking of sensitive information. Lastly, the goal of developing and implementing predictive analytics in child protection is not to replace the role of professionals but rather to help broaden the coverage of children at risk by providing an initial screening.

The remainder of the study is organized as follows. Section 2 describes the phenomenon of violence against children in more detail. Section 3 provides a literature review on the use of predictive risk modeling (MPR) in child protection. Section 4 describes the data and the preparation of the dataset, as well as the methodology used. Section 5 presents the results and an analysis of the decisions regarding the use of the model. Finally, Section 6 concludes.

2. The phenomenon of violence against children

Violence against children can take many forms: neglect or negligent treatment, psychological violence, physical violence, corporal

punishment, sexual abuse and exploitation, torture, inhuman or degrading treatment, peer violence, self-harm, harmful practices, violence in the mass media, violence through information and communication, and institutional and systematic violations of children's rights. The focus of this study is on violent discipline (which includes both physical punishment and psychological aggression) and, more specifically, on corporal punishment.

In Latin America and the Caribbean, 63 % of children under the age of 15 experience violent discipline (psychological or physical) by an adult member of the household, and 43 % are subjected to corporal punishment (UNICEF, 2020). Unfortunately, Argentina is in line with the region: the percentage of children under 15 years of age who are subjected to violent discipline at home is 59.4 % (MICS, 2019–20).

Exposing children to violent discipline has several detrimental outcomes that affect their development and well-being in both the short and long term (UNICEF, 2017; Cuartas et al., 2019; Chakrabarti et al., 2020). Moreover, exposure to violence in the first years of life can affect the structure of the brain and cause lifelong impairments in cognitive and emotional capacities, in addition to contributing to high-risk and anti-social behaviors (Cuartas, 2019 in ECLAC – UNICEF, 2020). The negative effects on these capacities and skills can, in turn, affect the child's learning processes, academic achievements, graduation rates, physical and mental health, risky behaviors, and even income, since the aforementioned capacities are important for these processes (Moffitt et al., 2011).

In Argentina, there are no publicly available national statistics on the number of child protection referrals. Only the capital city publishes annual statistics (Statistical Yearbooks of the Council for the Rights of Children and Adolescents of the City of Buenos Aires). The other sources of information are specific referral channel or program (e.g., reports on calls to the telephone 102 hotline, or on the “*Las Violencias*” program).

Data from the 2019–2020 MICS survey show that in the Buenos Aires metropolitan area, 3.9 % of children experienced severe corporal punishment. Extrapolating this prevalence to the population of children in the city of Buenos Aires³ suggests that approximately 27,000 children experienced severe physical punishment in the month preceding the survey. By contrast, in 2019, the interventions initiated in the city of Buenos Aires involved only 11,774 boys and girls (Buenos Aires CDNNyA Statistical Yearbook, 2019). Of these interventions, 61.8 % were related to the protection of children against violence, of which 29 % were related to physical abuse. The difference between the estimated prevalence of severe corporal punishment and the cases treated is an indication of the high levels of underreporting of this phenomenon.

3. Risk assessment in child protection

In the field of child protection, two primary approaches have traditionally been employed to assess a child's risk of experiencing violence and inform decision-making processes: theoretical or consensus-based models and actuarial models (D'Andrade et al., 2008). Theoretical or consensus-based instruments rely on theories of child maltreatment, related literature, and expert practitioner experiences to identify risk factors and develop assessment models (Cuccaro-Alamin et al., 2017). However, they have been criticized for subjectivity, inconsistency, and the use of similar variables to predict different outcomes (D'Andrade et al., 2008). Conversely, actuarial models incorporate statistically validated risk factors associated with future maltreatment, assigning weights to each factor to generate a score for individual cases (D'Andrade et al., 2008; Cuccaro-Alamin et al., 2017).

³ This assumes that the prevalence of child severe physical punishment is the same in all the localities of the Buenos Aires metropolitan region. Information on the child population in Buenos Aires City was retrieved from: <https://www.buenosaires.gob.ar/sites/gcaba/files/situaciondelasinfanciasyadolescenciasenlacaba2015-20191.pdf>.

² This work stemmed from the Caring Project (the real name is not included for confidentiality reasons), an initiative that incorporates predictive analysis in the implementation of public policies in Argentina.

In recent years, a novel approach known as Predictive Risk Modeling (PRM) has emerged in child protection. PRM utilizes predictive analytics to identify patterns and make predictions based on comprehensive data analysis (Cuccaro-Alamin et al., 2017). Unlike traditional methods, PRM considers a wide range of characteristics rather than relying solely on established risk factors from literature or practitioner experience. Successful implementation of PRM requires access to a broad target population, abundant and regularly updated data on risk factors, and clear legal and ethical protocols for prediction utilization (Vaithianathan et al., 2013).

PRM offers several advantages, including the ability to uncover previously unidentified relationships between variables and the flexibility to adapt models to reflect emerging data trends (Cuccaro-Alamin et al., 2017). Although PRM has been widely utilized in healthcare for predicting adverse outcomes, its application in child protection is relatively recent.

Vaithianathan et al. (2013) pioneered the application of PRM in child protection, utilizing a generalized linear model with administrative data from New Zealand to estimate the likelihood of substantiated maltreatment by age 5. Following a preventive approach, Chile's Ministry of Social Development and Family developed the "Sistema Alerta Niñez," using administrative data to predict children's risk of rights violations, aiding in preventive services and family support. Meanwhile, initiatives like the Allegheny Family Screening Tool (AFST) and the Eckerd Rapid Safety Feedback Process (ERSF) in the United States demonstrate PRM's effectiveness in prioritizing child maltreatment referrals and assessing the risk of life-threatening episodes for children in the protective system, respectively. Furthermore, recent studies by Amrit et al. (2017) and Clayton et al. (2020) have explored the use of structured and unstructured data in predictive analytics for child protection. These studies aim to predict child maltreatment and support social workers in their decision-making processes.

Notably, Argentina has also witnessed pioneering efforts in predictive analysis for child protection, such as the Child Labor Risk Identification Model developed by the International Labour Organization and the Economic Commission for Latin America and the Caribbean. This model utilized various data sources to identify areas with high child labor risk and inform targeted interventions (ILO-ECLAC, 2018).

4. Data & methodology

In this section, we first present the data used in the analysis. Second, we present the models used to predict child physical violence. We consider two strategies: (i) random forest; and (ii) lasso-regularized logit regression (lasso-logit from now on).⁴ Third, we will explain in detail the pipeline that is followed for the proper use of predictive models implemented in this case. Next, we present the measures used to evaluate the performance of the models. Finally, we explain the criteria for selecting the thresholds used to assign the observations to a class.

4.1. The data

The effectiveness of the machine learning models to predict the presence of physical punishment of children at the household level hinges on access to extensive information about the child and their environment. While such data is likely available within Argentina's child protection system administrative records, it is not publicly accessible. Therefore, we utilize the Multiple Indicator Cluster Survey (MICS), conducted by UNICEF in Argentina mainly for illustrative purposes, showcasing how this data can extend the reach of early detection efforts. Additionally, our work underscores the importance of integrating information that may be scattered across public agencies, even if it is not

immediately suitable for this purpose. Furthermore, this exercise may reveal gaps in existing data collection practices and highlight areas where additional data collection efforts are needed.

The MICS conducted by UNICEF in Argentina in 2019–20 is a nationally representative survey consisting of four questionnaires: (i) a household questionnaire; (ii) a questionnaire for women aged 15–49; (iii) a questionnaire for children under 5 years of age, administered to mothers (or caregivers) of all children under 5 years living in the household; and (iv) a questionnaire for children and adolescents aged 5–17, administered to the mother (or caregiver) of a child aged 5–17 living in the randomly selected household.

Of the 14,968 households included in the MICS survey, we restricted the sample to households with children aged 1–14, as these were the households eligible for the child discipline module that we used to construct the outcome of interest. As a result, our sample consists of 7,358 households.

The child discipline module contains relevant information to develop models that are able to predict the presence of physical punishment of children. Mothers or caregivers of children under five years of age and of a randomly selected child aged 5–17 were asked a series of questions about the methods used by adults to discipline a child at home during the previous month. According to the MICS definition of physical punishment of children,⁵ a dummy variable was created, that takes the value 1 if the caregiver reported that he or she or someone else in the household used any form of physical punishment against the child (in the month prior to the MICS interview), and 0 otherwise.⁶

The variables included in the analysis can be grouped into the following themes: socioeconomic indicators, working conditions, sanitary and health coverage, housing infrastructure, and child development. In the process of preparing the dataset, some categorical variables were one-hot-encoded, and of some categorical variables and some individual variables were grouped at the household level. The final database contains 643 variables. In addition, for those variables with missing values, imputation was performed by replacing these missing values with the conditional median of the variable in each wealth quintile,⁷ as reported in the database. It is important to emphasize that most variables either had no missing values or less than 1 % of observations were missing and that these were uniformly distributed across the quintiles.

The nature of the data may be cause of concern. First of all, self-reporting may lead to underreporting. Furthermore, it could be the case that some particular groups show larger underreporting rates, biasing the predictions. For example, if, due to cultural factors or social pressure, households of a certain socioeconomic level are more likely to believe that it would be frowned upon to use violent methods to discipline their children, then underreporting may be higher in this group than in others. As a result, the predicted use of violent discipline methods in such groups may be lower than it is. Unfortunately, neither general underreporting nor the potential bias may be solved by statistical procedures. However, it is worth noting that the number of people who reported the use of violent disciplinary methods in the MICS 2019–20 is significant, suggesting that at least a high proportion is reported. And although reporting rates are not uniform across socio-economic groups, they are relatively high for all of

⁵ <https://data.unicef.org/topic/child-protection/violence/violent-discipline/>

⁶ An alternative way of representing child physical punishment was explored. An ordinal variable was created to account for the severity or intensity of physical punishment. In this case, the variable took the lowest value if physical discipline methods were not used and higher values depending on the number of severe and non-severe physical punishment methods used. Multinomial models were estimated using this result, but the predictive performance was poor.

⁷ For instance, for a missing variable X belonging to quintile q , the imputed value was the median of X across quintile q .

⁴ A CART model was also estimated, but, as expected, its performance is poor compared to the other models.

them, ranging from over 52 % for the lowest quintile to almost 30 % for the richest one. This reassures the use of this data, given that there is no other publicly available source of information on child violence at the national level in Argentina.

4.2. Models and implementation

We will analyze two approaches to the supervised learning problem. On the one hand, we consider an ensemble model based on classification trees, random forest, which is a non-parametric approach characterized by its flexibility. On the other hand, we consider a lasso-regularized logistic regression, which is characterized by its interpretability. Both methods predict the probability of child violence in a household. The main difference between them is that the former is a black-box procedure but explores nonlinearities among the regressor variables, while the latter is a crystal-box model but does not consider nonlinear relationships. For a more detailed explanation of these procedures, see the Modeling Methods section in the Appendix.

One of the main challenges in training a learning model is to keep the bias and variance of the models low. The literature shows that the standard approach to address this problem is to divide the sample into two random subsets, the training sample, which consists of 70 % of the observations, and 30 % will be the validation set.

To have a model that compensates for bias and variance and has good predictive power, the hyperparameters of each model were trained on the training sample using a *k*-fold cross-validation procedure. In this procedure, we randomly divide the training sample data into *k* disjoint subsamples of equal size. At each step, the model is fit using *k* – 1 of the subsamples of the data; the remaining subsample is used to measure the performance of the model. We repeat this procedure over the *k* subsamples and calculate the average misclassification error of the predicted results. The subset of parameters with the best average performance is selected.

In the case of random forest, the parameter chosen for cross-validation was the number of regressors considered in each partition of each tree, following a 5-fold cross-validation scheme with two replicates, and the metric used was ‘accuracy’, which we will describe in the next section. For the lasso-logit model, the optimal parameter was selected by a 10-fold cross-validation scheme and λ was chosen so that the cross-validated error is within one standard error of the minimum that yields to the most regularized model.

4.3. Metrics for evaluating model performance

One of the most important characteristics of classification problems is that different types of errors can be made when misclassifying observations. These errors are not symmetric, and therefore it is relevant to describe them in order to understand the implications of committing each of them, which are shown in Table 1. The confusion matrix (Table 1) shows the predictions and allows comparison with the actual

state. The “positives” should be interpreted as the households in which physical punishment is used and therefore we try to identify them. The “negatives” refer to households without physical punishment. There are four possible outcomes: correct classifications, which are true positives and true negatives, and misclassifications, which are false positives (predicting violence when there is none) and false negatives (not predicting violence when there is violence).

The confusion matrix is a contingency table that allows the computation of a set of measures describing the model fit. It is used as a starting point for constructing metrics that focus on the errors to be avoided in the context of each problem studied. The first one is the sensitivity (also called the “true positive rate” or “recall”), which is the proportion predicted to be violent among those who are, TP/P . The second one is the specificity (also called “true negative rate”), which is the proportion of cases that are predicted to be nonviolent out of the total number of cases in which violence is not recorded, TN/N . Complementary to these measures, the false negative rate, FN/P , and the false positive rate, FP/N , can be calculated. In addition, the accuracy is the total proportion of correctly predicted observations, i.e., $(TP + TN)/(P + N)$.

In some contexts, positive predictive value or precision is also a relevant measure. Positive precision (PPV) measures how many cases predicted to be positive (with violence) are positive, TP/P^* . Similarly, the negative predictive value (NPV) measures how many cases predicted to be negative are negative, i.e. TN/N^* .

The models used in this paper produce a score for each household ranging between 0 and 1. In both cases, the models predict the probability of physical punishment, high output values imply a prediction of a violent household, while low values imply a prediction of a non-violent household. Then, in each case, it is crucial to determine a threshold at which an observation is classified as violent. Once this threshold is set, the confusion matrix is constructed, and the predictive value of the model can be analyzed.

4.4. Selecting a threshold for predicting physical violence

The procedures used in this paper, random forest and lasso-logit, assign each observation a score between 0 and 1 that indicates the risk of child violence in that household. The classification of households with child violence is determined by setting a threshold above which a family is classified as having this characteristic. Both sensitivity and specificity are functions of the thresholds. One way to determine the cut-off threshold while taking into account sensitivity and specificity is to use Youden’s (1950) index, *J*, which is given by

$$J = sensitivity + specificity - 1$$

The index is calculated on the training sample; it takes values between 0 and 1, with 1 being the preferred value, when there are no false positives or false negatives cases. Then, the selected threshold is determined by the maximum of *J*, and it coincides with the point of the ROC curve that minimizes the distance to the point (0, 1). The index favors sensitivity and specificity indicators, and it is useful to choose an optimal threshold to formulate predictions in the test set.

It should be noted, however, that if other performance metrics are preferred, a different threshold and criterion could be chosen. To address this issue, in Section 5.2.2 we present a sensitivity analysis that varies the threshold of the classifiers, considering values close to the chosen threshold using the Youden index. On the other hand, taking into account that the aim is to design a model that could eventually be implemented, a further operational limitation is the capacity of the Argentinian protection system to deal with cases of children at risk. Given this constraint, the problem of choosing the threshold could be solved implicitly by ordering households according to their predicted score and prioritizing those with the highest scores until the maximum number of households that can be attended is reached. This criterion will be referred to as “maximum household reach”. In Section 5, we present

Table 1
Confusion matrix.

	Predicted Class		Total
	Negative	Positive	
	(without violence)	(violence)	
	Negative	True Negative (TN)	
	Positive	False Positive (FP)	
Real Class	(without violence)		Negatives (N)
	Positive	False Negative (FN)	Positives (P)
	(violence)	True Positive (TP)	
Total	Predicted Negatives (N*)	Predicted Positives (P*)	

the results considering a hypothetical constraint where only 30 % of the households could be reached, which seems to be a reasonable percentage in the context in which the research is proposed. Results are robust to alternative constraints, as shown in Table A.1 in the Appendix.

5. Results

This section first describes the phenomenon of physical punishment of children. Second, the results of the models are presented, contrasting the performance of different models and the implications of using thresholds chosen according to the Youden index criterion and the “maximum household reach” criterion. Third, an analysis of the bias of the model is provided. Fourth, we study which features are central for predicting accurately violence against children. Finally, the ability of the models to identify severe physical violence is examined.⁸

5.1. Descriptive statistics

The data set under analysis shows that in 41.8 % of the households, caregivers use at least one physically violent method to discipline their children. However, when disaggregating the use of physical punishment according to the methods used, we observe differences in their prevalence (Fig. 1). The most commonly used method is “slapping on the bottom with a bare hand”, reported by 23.6 % of the households, followed by “shaking”, present in 22.2 % of households. In contrast, “beaten up”, i.e., repeatedly hitting the child as hard as possible, and “hitting with a hard object” (belt, brush, stick, or other hard objects) are the least common methods, with prevalence rates of 1.2 % and 4.6 % respectively. Furthermore, according to the MICS definition (UNICEF, 2020), “beating up” and “slapping on the face or hitting on the head or ears” are considered as severe physical punishment, the latter being more common in Argentine households.

Furthermore, when examining the correlation between the various methods of physical punishment surveyed in the MICS, we note that, in general, these correlations are relatively low (see Figure A.1 in the Annex). This fact might suggest that considering the different methods to construct the outcome to be predicted is informative and not necessarily repeated information. The methods “hitting the arm, hand or leg” and “Slapping on the bottom with the bare hand” show the highest correlations, which could be partly explained by their higher prevalence. While “Beaten up” is the least correlated method with the rest.

In addition, it is relevant to examine the prevalence of physical punishment in different socioeconomic groups. Fig. 2 shows the prevalence of violent physical discipline among children in five wealth quintiles.⁹ The poorest quintile shows the highest prevalence of physical punishment, and this rate decreases for the richest quintile, which shows the lowest prevalence.

5.2. Model results

According to our random forest and lasso-logit models the proportion of households where children are disciplined based on physical

violence is 36 % and 48 % respectively (see Panel A in Table 2). This classification is based on the optimal threshold selected using the Youden index, which is 48 % in the case of random forest and 40 % for lasso-logit. This means that if the probability estimated by the random forest model (resp. lasso-logit) is greater than or equal to 0.48 (resp. 0.40), then a household is classified as violent.

Panel B in Table 2 provides a series of performance indicators that allows for comparing both models. In the first place, the more general measures show a consistent pattern across models. In fact, accuracy is similar for both models, which make a correct prediction for 63 % of the observations of the sample.

In terms of sensitivity and specificity, models show larger differences. Out of the total number of households using violent methods, random forest is able to identify 49 % correctly, compared to 63 % of lasso-logit. In contrast, when focusing on households that do not use violent methods, random forest outperforms lasso-logit: while the former classifies correctly 74 % of the sample, the latter does so in 63 % of the observations.

Finally, precision is similar across models. Out of ten households predicted to be using violent methods for disciplinary purposes, almost six of them actually used them. Conversely, random forest has a 66 % negative precision value close to the 69 % rate of lasso-logit, implying that almost 7 out of the 10 households that were predicted not to be using violent methods were actually not using them.

5.2.1. Sensitivity analysis

The performance of the indicators depends on the threshold chosen. In this paper, we propose to follow the Youden index to set it and obtain a threshold of 40.1 % (resp. 48 %) for lasso-logit (resp. random forest). This section assesses how sensitive both models are to changes in this threshold.

Table 3 shows the performance of lasso-logit under five different thresholds (including 40.1 %, the one proposed in this paper, in bold font).¹⁰ As expected, the lower the threshold chosen, the higher the proportion of households that are predicted to rely on violent methods across the sample: from 65 % based on a threshold of 35 % to 15 % when it is set to 55 %. Noticeably, accuracy does not virtually change across different thresholds. Indeed, for all thresholds shown, the model is able to predict correctly (as users and not users of violent methods) between 60 and 64 households out of 100. In contrast, sensitivity and specificity change sharply across thresholds. As expected, the lower the threshold, the higher the sensitivity and the lower the specificity. That is, as we lower the threshold, we are able to identify more households out of those that do use violence but at the risk of reducing the proportion of those that do not use violence and are classified in that way. In short, if the aim is to capture as many positive cases as possible, but at the cost of a higher false positive rate, a lower threshold should be used.

Table 4 provides results derived from choosing different thresholds in the case of random forest, including the threshold derived from the Youden Index, which is the one proposed in this paper (in bold font). By construction, the proportion of households predicted to use violent methods is higher the lower the threshold (see Panel A). Similar to lasso-logit, accuracy shows almost no changes, ranging from 60 to 63 % across different thresholds. Also, sensitivity decreases as thresholds increase, while, as expected, the opposite is true for specificity. Conversely, in positive precision tends to increase with thresholds while negative precision decreases. Noticeably the range of values for all four

⁸ Statistical analysis was performed using the free software R Core Team (2021). The code and database are available for reproduction at https://git-front.io/r/user-3641133/ZKGgzuob167w/child-protection/blob/Models_MICS19_20.R.

⁹ The definition of wealth used is that provided by the MICS survey: a composite wealth indicator based on information on ownership of consumer goods, housing characteristics, water and sanitation, and other characteristics related to household wealth. Principal component analysis is used to generate the weights of the variables. Each household is assigned a wealth score, and the population is classified and divided into five wealth quintiles, from the lowest (poorest) to the highest (richest) (see National Secretariat for Children, Adolescents, and Family and United Nations Children's Fund, 2013).

¹⁰ While predictions resulting from a large number of thresholds could be considered, in Table 3, we constrain ourselves to predictions using thresholds close to the one chosen according Youden index. These results classified as positive between 15% and 65% of the households of the validation sample. The reason for this decision is that, in practice, there would be restrictions that make it difficult to analyze and address the situation of all the households.

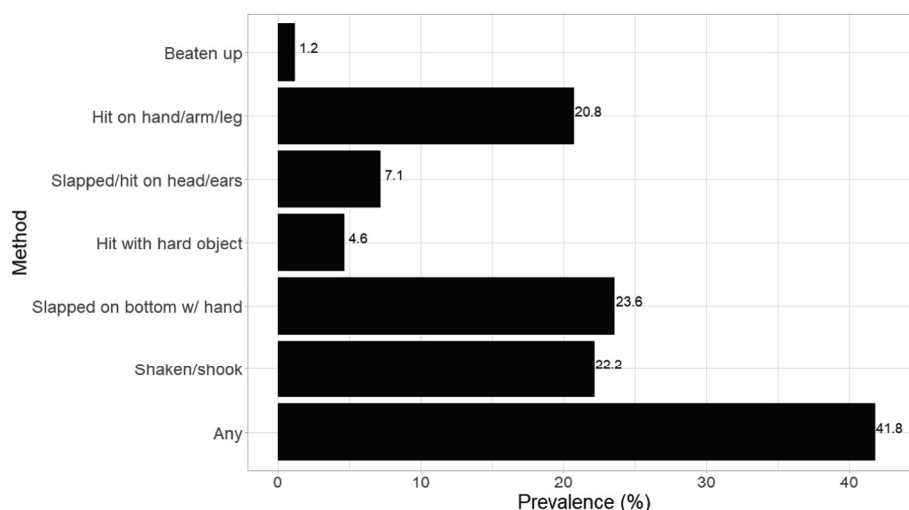


Fig. 1. Prevalence of physical punishment, by the method used.
Source: own elaboration based on MICS survey of Argentina (2019-20)

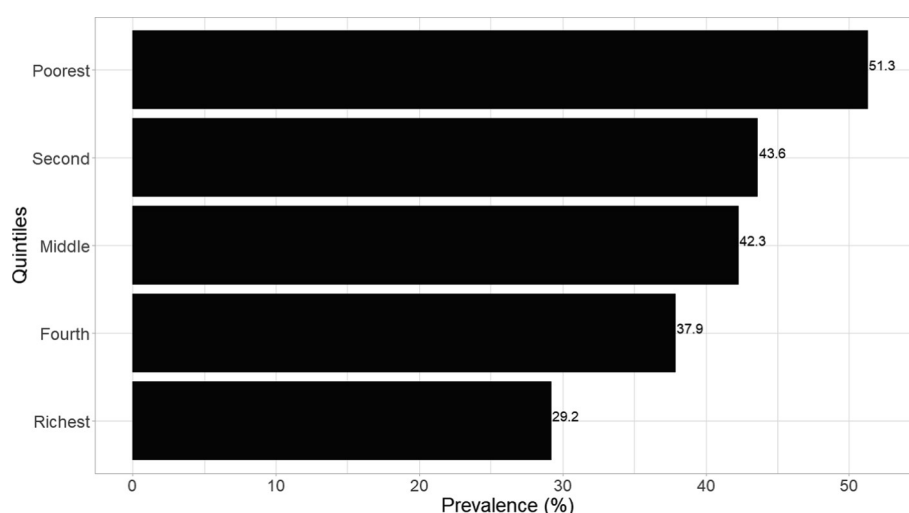


Fig. 2. Prevalence of physical punishment, by wealth quintile.
Source: own elaboration based on MICS survey of Argentina (2019-20)

Table 2

Prediction, thresholds and performance, selected threshold via the Youden Index.

	Random Forest	Lasso-logit
<i>Panel A: prediction and thresholds (%)</i>		
Threshold	48	40.1
P*/Total	36	48
<i>Panel B: prediction performance (%)</i>		
Accuracy	63	63
Sensitivity	49	63
Specificity	74	63
Positive precision (PPV)	59	57
Negative precision (NVP)	66	69

Source: own elaboration based on MICS survey of Argentina (2019–20).
Note: P*/Total is the proportion of predicted positive cases over the total number of observations in the training sample.

performance indicators (sensitivity, specificity and positive and negative precision) are similar across models.

These results suggest that public policy constraints and the tolerance for one type of error or another could inform the extent to which a lower

Table 3

Performance of the lasso-logit classifier for different thresholds.

<i>Panel A: prediction and thresholds (%)</i>					
Threshold	35 %	40.1 %	45 %	50 %	55 %
P*/Total	65	48	32	22	15
<i>Panel B: prediction performance (%)</i>					
Accuracy	60	63	64	63	62
Sensitivity	79	63	45	33	23
Specificity	46	63	78	86	91
Positive Precision (PPV)	53	57	60	65	66
Negative Precision (NPV)	75	69	65	63	61

Source: own elaboration based on MICS survey of Argentina (2019–20).

or higher threshold is needed. In this regard, in what follows we show the results when considering a logistic or budgetary constraint (which could appear in a public policy setting).

Table 5 shows the performance indicators of the classifiers when the constraint implies that the maximum percentage of households that can be served is 30 % due, for instance, to logistical constraints (i.e., the total number of predicted positive cases should not exceed 30 % of the households). In this case, the predictions are ranked according to their

Table 4
Performance of the random forest model for different thresholds.

Panel A: prediction and thresholds (%)					
Threshold	40 %	45 %	48 %	50 %	55 %
P*/total	57	43	36	28	18
Panel B: prediction performance (%)					
Accuracy	60	62	63	63	63
Sensitivity	70	56	49	40	27
Specificity	53	67	74	81	89
Positive Precision (PPV)	53	56	59	61	66
Negative Precision (NPV)	70	67	66	64	62

Source: own elaboration based on MICS survey of Argentina (2019–20).

Table 5
Prediction performance, threshold selected via the criterion “maximum household reach”.

	Random Forest	Lasso – logit
Panel A: prediction (%)		
P*/Total	30 %	30 %
Panel B: prediction performance (%)		
Accuracy	63 %	64 %
Sensitivity	42 %	43 %
Specificity	79 %	80 %
Positive Precision (PPV)	60 %	62 %
Negative Precision (NPV)	64 %	65 %

Source: own elaboration based on MICS survey of Argentina (2019–20).

score, and 30 % of the households are determined to be positive. Both models show very similar performance, which might suggest that the structure of the problem is linear. On the other hand, the results are close to those obtained by random forest using the threshold given by the Youden index criterion. In this case, lasso-logit has a sensitivity of 43 %. The accuracy is 64 % and the precision is 62 %. Table A.1 in the Annex shows that results are robust when criteria other than 30 % are considered.

5.3. Feature importance

One of the main purposes of lasso-logit is to obtain sparse models by selecting the variables that contribute the most to the prediction of violence in our case. By applying this procedure, our final model depends on only 27 % of the original variables. Among these variables, we can point out that those with the greatest participation in the model correspond to issues related to housing infrastructure, integral health of household members, and education. These are followed by variables related to characteristics that describe the demographic composition of

the household (age and gender composition, whether they have living parents and live with them) and the education of household members (educational levels attained, school dropout). There are also socioeconomic indicators of the households (e.g., whether they receive transfers and their level of wealth). Finally, there are variables related to the education and stimulation of young children. A detailed description of the variables appears in the Appendix.

In the case of random forests in classification problems, we can obtain a measure of the importance of each predictor by averaging over all trees the reduction in the Gini index resulting from the partition in which each variable is involved. The most important variables are those that produce the largest reduction in the Gini index. Fig. 3 shows an elbow indicating that the first variables make larger contributions to the random forest model and then present a plateau with smaller contributions. The first 55 variables contribute to 38 % of the reduction in the Gini index. The variables are shown in detail in the Appendix. The results are consistent with those obtained using the lasso-logit model. Some groups of variables seem to have a relatively high importance. They are those related to the socio-economic status, the demographic composition, the access to health services, and the variables related to the child of the reading in the households.

Looking at the variables that explain up to 75 % of the sum in the reduction of the Gini index, we find those related to the infrastructure of the house and variables that identify the caregiver in charge (responsible adult or siblings). Finally, some variables related to the employment status, not only of the adults in the household but also of the adolescents in the family, have a relatively high importance.¹¹ Previous literature supporting the hypothesis of a link between stress and child maltreatment may be related to this last point (Jonson-Reid et al., 2009).

5.4. Biases

To complement the analysis, an additional exercise is to examine the performance of models in different wealth quintiles (see Section 5.1 for a definition of wealth). It would be desirable for the models to have the same metrics across all quintiles. Table 6 summarizes the performance of the lasso-logit model across five wealth quintiles (considering the predictions using the optimal threshold according to the Youden index, as in Table 2).

The model has significantly higher sensitivity values in the poorest quintile compared to the richest quintile. In addition, the accuracy is higher for the richest quintile. For the random forest model, the results are similar (Table 7). These results suggest that, if implemented, there would be a risk of possible stigmatization of the poorest quintiles and loss of at-risk children from the highest quintiles.¹²

It is therefore necessary to implement strategies to reduce this gap. One possible approach is to set different thresholds for different quintiles.¹³ So that the poorer quintiles have higher thresholds, thereby reducing the false positive rate in these groups. Similarly, it would be reasonable to set lower thresholds for richer quintiles, thereby increasing sensitivity. We explored this approach by setting thresholds

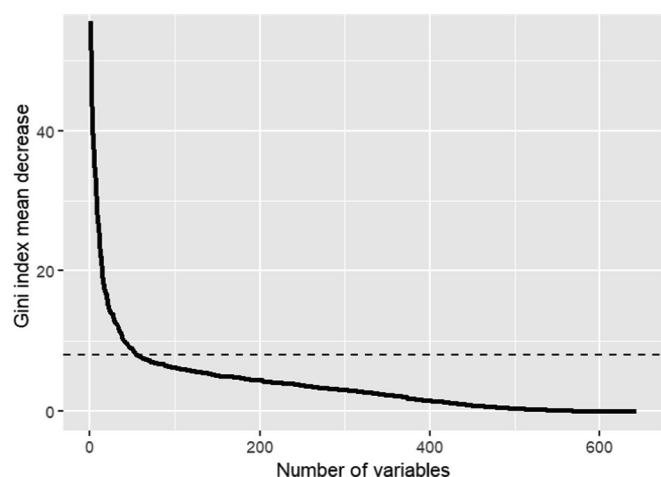


Fig. 3. Decreasing average Gini index ordered in descending order.
Source: own elaboration

¹¹ Detailed results for the two models are available on request from the authors.

¹² It would be interesting to compare this model with existing ones in terms of its bias. Although some literature suggests that the child protection system may be biased against lower-income families (Brown et al., 1998), the available data on physical violence against children in Argentina does not allow us to estimate this problem.

¹³ In a public policy scenario in Argentina, to identify families with different levels of wealth without having to ask directly about wealth or income, one could use some variable that reflects the socioeconomic status. For example, an alternative could be identifying households receiving the *Asignación Universal por Hijo (AUH)*. In this way, different thresholds could be determined for households that receive this conditional cash transfer and households that do not.

Table 6
Performance of the lasso-logit model on wealth quintiles (%).

	Q1 (Poor households)	Q2	Q3	Q4	Q5 (Rich households)	All households
<i>Panel A: prediction and thresholds (%)</i>						
Threshold	40.1	40.1	40.1	40.1	40.1	40.1
P*/Total	75.2	56.3	45.0	31.4	15.5	48.0
<i>Panel B: prediction performance (%)</i>						
Accuracy	60	62.1	62.6	61.5	72.5	63.2
Sensitivity	83.9	71.6	58.8	41.6	29.4	63.1
Specificity	34.2	55.1	65.6	76.1	90.3	63.1

Source: own elaboration based on MICS survey of Argentina (2019–20).

Table 7
Performance of the random forest model on wealth quintiles (%).

<i>Panel A: prediction and thresholds (%)</i>						
Threshold	48	48	48	48	48	48
P*/Total	63.0	41.7	30.0	20.4	8.0	35.7
<i>Panel B: prediction performance (%)</i>						
Accuracy	61.8	63.2	61.5	59.7	71.3	63.2
Sensitivity	73.8	55.8	40.2	26.5	14.7	48.7
Specificity	48.7	68.8	77.9	84.1	94.7	74.1

Source: own elaboration based on MICS survey of Argentina (2019–20).

of a heuristic nature for one of the two models (lasso-logit).¹⁴ For this exercise, we considered two groups and chose different thresholds for each (Table 8). The first group consists of households belonging to the two poorest quintiles, and the second group consists of households belonging to the three richest quintiles.¹⁵ If we set a threshold that is 5 percentage points higher than the threshold selected by the Youden index for the first group, and 5 percentage points lower for the second group, we obtain the following metrics:

This approach allows us to obtain almost the same overall true positive rate while reducing the gap between the poorest and richest quintiles (at the expense of a slightly higher overall false positive rate).

5.5. Models' ability to identify severe physical punishment

In the procedures analyzed, the outcome to be predicted was whether any physical punishment methods were used to discipline children, regardless of the severity of the method used. It could be argued that children living in households where severe physical punishment is used are at greater risk.¹⁶ Therefore, it is desirable to analyze the ability of the models to identify households where severe physical punishment methods are used. Table 7 compares the true positive rates for physical punishment, defined as moderate and/or severe physical punishment, and the true positive rates for severe physical punishment alone, for the lasso-logit model predictions obtained with different thresholds.

Given that the prevalence of severe physical punishment is lower than that of moderate punishment, it may be the case that the model has difficulty in finding these households. However, the results in Table 9 show that the lasso-logit model captures households where severe

¹⁴ We also explored this approach with the random forest model (see Table A.2 in the appendix). Results are similar.

¹⁵ As mentioned in the previous footnote, in a public policy scenario in Argentina, identifying five wealth quintiles may be a difficult task. However, identifying two economic groups seems more realistic and feasible to achieve, for example, considering an income proxy variable, such as being a beneficiary of the AUH.

¹⁶ In fact, as an exploratory exercise, models using severe physical violence –without sampling using balancing techniques– were estimated. Their performance was poor, probably due to the imbalance of the two categories. An extension of this work could be to explore the use of class balancing techniques to determine whether or not classification performance can be improved.

Table 8
Metrics for the lasso-logit model, thresholds specific to socio-economic features.

	Q1 (lowest)	Q2	Q3	Q4	Q5 (Highest)	All households
<i>Panel A: prediction and thresholds (%)</i>						
Threshold	45	45	35	35	35	48
P*/Total	57.8	35.2	66.4	52.8	27.8	49.3
<i>Panel B: prediction performance (%)</i>						
Accuracy	60.0	64.3	61.3	59.4	69.9	62.6
Sensitivity	67.1	49.5	82.0	64.5	46.1	63.9
Specificity	52.4	75.4	45.5	55.8	79.8	61.7

Source: own elaboration based on MICS survey of Argentina (2019–20).

Table 9
Comparison of the sensitivity of physical punishment and severe physical punishment for the lasso-logit model.

Panel A: prediction and thresholds (%)					
Threshold	35 %	40.1 %	45 %	50 %	55 %
P*/Total	65	48	32	22	15
Panel B: prediction performance (%)					
Sensitivity to physical violence	79	63	45	33	23
Sensitivity to severe physical violence	69	69	55	47	34

Source: own elaboration based on MICS survey of Argentina (2019–20).

Table 10
Comparison of the sensitivity of physical punishment and severe physical punishment for the random forest model.

Panel A: prediction and thresholds (%)					
Threshold	40 %	45 %	48 %	50 %	55 %
P*/Total	57	43	36	28	18
Panel B: prediction performance (%)					
Sensitivity to physical violence	70	56	49	40	27
Sensitivity to severe physical violence	75	64	53	50	39

Source: own elaboration based on MICS survey of Argentina (2019–20).

physical punishment is used to discipline children. In fact, when we focus on the household where severe punishment is used, the sensitivity of the model is even higher than that calculated when considering severe and non-severe physical violence, except when the threshold is 35 %. These results provide an additional validation of the model's performance. The random forest model is also capable of identifying severe physical violence, showing higher sensitivity for this outcome. Table 10.

6. Conclusions

Physical violence against children is a complex and pressing problem, with high levels of underreporting. This study attempts to show the feasibility of developing models based on machine learning techniques. The suggested predictive models could expand early detection coverage by identifying children who might be at risk, thus addressing underreporting and bias concerns. Specifically, these models could help pinpoint factors that could be incorporated into the initial screening of at-risk children.

The results show that the models have the potential to identify a substantial proportion of households where physical punishment methods are used. The results obtained with both techniques are consistent. In this particular case, the lasso-logit model performs slightly better. When we use the threshold selected according to the Youden index criterion for prediction, the lasso-logit classifies 48 % of the sample as households where physical punishment is used, obtaining a sensitivity and a specificity of 63 %. Random Forest classifies 36 % of households as using violent methods, with a sensitivity of 49 % and a specificity of 74 %.

If instead we consider a constraint that limits the number of households predicted as positive to 30 % of the sample, lasso-logit and random

forest models are almost identical, achieving a sensitivity of 43 % the former and 42 % the latter, with a specificity of 80 % and 79 % respectively.

In terms of public policy, these results imply that the proactive approach could be an improvement over the existing status quo in child protection. According to international estimates, self-reported physical abuse by children is estimated to be 75 times higher than what official records indicate (Stoltenborgh, et al., 2013). Through the use of predictive models in child protection, assessing of the situation of 30 % of families could lead to the early detection of 43 % of households where children experience physical violence.

However, it is crucial to consider the risks of using predictive analytics in this setting. The models' predictions are not always accurate and false positives could stigmatize families. Careful implementation could minimize some of the model's weaknesses. In particular, we suggest using the models for identification facilitating the initial screening of potential victims, which should then be adequately assessed by professionals trained to perform that task. Privacy must also be guaranteed throughout the implementation. In addition, a major challenge is to reduce the model's bias between income quintiles. The use of different thresholds according to socio-economic groups would address this issue.

As mentioned earlier, these models could likely perform well using administrative data, which should be amalgamated to incorporate a diverse range of variables into the models and evaluate the significance of various predictors. Lacking access to this data, we rely on the Argentinian 2019–20 MICS survey to develop the model for initial screening. It is important to note, however, that the analysis proposed using MICS data could be valuable not only as an illustration, but also for its ability to point to predictors that should be included when using

alternative data. For instance, variables featuring the household composition, educational level, health indicators and dwelling characteristics. Moreover, our study shows that is important to retrieve information related to daily habits, such as whether children have access to books, whether they are stimulated at an early age, and so on.

These and other strategies need to be further explored. Future research should focus on improving the performance of the models to strengthen some of their weaknesses, such as the over-representation of households from the lowest quintiles. In addition, models could be developed to adequately capture the phenomenon of violence against children and feed into an early warning system to identify children at risk.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data has been uploaded to github

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Appendix

Modeling Methods

Random forest (Breiman, 2001) is a supervised learning procedure based on ensembles, characterized by its high predictive power. This procedure is constructed based on several classifiers with high variance, in this case, classification trees (Breiman et al., 1984). The predictions of each tree are merged into a single prediction by majority or weighted voting. In this way, we obtain a classifier that combines the low bias of the initial estimators with the low variance of an estimator assembled by the majority or weighted voting. An advantage of this approach is that it allows the study of problems where the relationship between the features and the response variable is highly nonlinear and complex. Although this is a black-box procedure, its output is rich since it reports the estimated probability of a given observation being classified as violent, and a descending scale of the importance of the variables, which allows for determining the incidence of the variables in the classification. In this way, we also gain interpretability of the problem. The algorithm depends on several parameters, such as the number and depth of the initial trees. The parameters are determined by cross-validation and are explained later in the text. For more details on classification trees and random forest see Hastie et al. (2009).

Logistic regression is a widely used classifier that models the probability that the response belongs to a category. In this case, it models the probability that an observation corresponds to a household where child violence occurs. Based on these probabilities, we can estimate a threshold to determine whether an observation corresponds to the analyzed category, in this case, the presence of child violence.

In binary problems, such as the one we are analyzing, the modeled response probability follows the following equation:

$$p = P(Y = 1|X = x) = F(\beta_0 + x'\beta),$$

where F is the logistic link function $F(z) = \frac{\exp(z)}{1 + \exp(z)}$, Y is the binary response variable, and X is the vector of regressor variables. Finally, (β_0, β) is the vector of $(p + 1)$ parameters estimated by maximum likelihood, given that $Y|X = x \sim \text{Bi}(1, p(x))$. That is,

$$(\hat{\beta}_0, \hat{\beta}) = \text{argmax}_{(\beta_0, \beta)} l(\beta_0, \beta) = \text{argmax}_{(\beta_0, \beta)} \sum_{i=1}^n y_i (\beta_0 + x'_i \beta) - \log(1 + \exp(\beta_0 + x'_i \beta)),$$

where $l(\beta_0, \beta)$ is the log-likelihood function. As with linear regression models, the lack of parsimony in logistic models leads to overfitting of the model, resulting in high variability in predictions, which leads to poor predictions for data outside the sample on which the fit was performed. On the other hand, the logistic model has the advantage of being a glass-box model with high explanatory power. In practice, this means that simple classification rules can be constructed and used by the end-user of the model, following colloquial rules that do not involve equations. Variable selection and regularization techniques are used to reduce the variance with a small increase in model bias. The results are parsimonious models, which, in turn, recover the ability to be easily interpretable. We focus on regularization models with l_1 -penalty, lasso type. This type of penalty allows some coefficients to shrink to zero and reduce the variance of the model. In practice, the vector is obtained by minimizing the following equation,

$$(\hat{\beta}_0, \hat{\beta}) = \operatorname{argmax}_{(\beta_0, \beta)} l(\beta_0, \beta) + \lambda \sum_{j=1}^p |\beta_j|$$

where the log-likelihood equation in the classical model is penalized by controlling the l_1 - norm of the coefficients accompanying the regressor variables, and finally λ controls this penalty. Later we will show how this parameter is estimated in a data-driven manner through cross-validation. For more details on these models see [Hastie et al. \(2009\)](#).

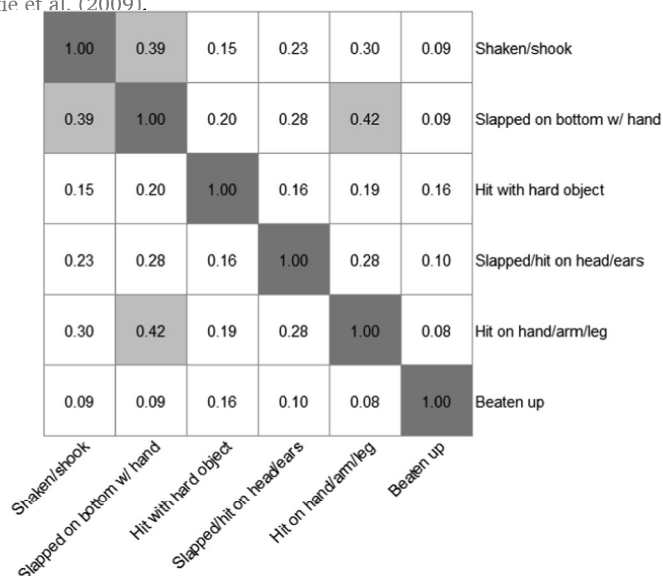


Fig. A1. Correlation matrix for physical punishment method. Source: own elaboration based on MICS survey of Argentina (201920).

Table A1

Prediction performance, threshold selected via the criterion “maximum household reach” (%).

	Random Forest	Lasso-Logit	Random Forest	Lasso-Logit	Random Forest	Lasso-Logit
P*/total	20	20	42	42	50	50
Sensitivity	30	30	55	55	63	65
Specificity	88	88	68	68	60	61
False Negative Rate	70	70	45	45	37	35
False Positive Rate	12	12	32	32	40	39
PPV (precision)	65	65	57	57	54	56
NPV	62	63	67	67	68	70
F1 score	41	41	56	56	58	60
Accuracy	63	63	63	63	61	63

Source: own elaboration based on MICS survey of Argentina (2019–20).

Table A2

Metrics for the random forest model, thresholds specific to socio-economic features.

	Q1(lowest)	Q2	Q3	Q4	Q5 (Highest)	All households
<i>Panel A: prediction and thresholds (%)</i>						
Threshold	43	43	53	53	53	
P*/Total	76.4	58.3	17.2	9.2	3.2	37.2
<i>Panel B: prediction performance (%)</i>						
Accuracy	59.2	61.4	61.3	59.7	71.6	62.1
Sensitivity	84.2	73.2	25.3	13.3	6.9	49.3
Specificity	32	52.7	88.9	93.8	98.4	71.8

Source: own elaboration based on MICS survey of Argentina (2019–20).

Variable selection random forest

In the following, we present the 55 variables that make the greatest contribution to the random forest model, grouped according to the categories: household composition, health, education, work, welfare, and lifestyle.

Demographic composition of the household.

- Average age of household

- Age of household head
- Mean age of children aged 0–17
- Mean age of children aged 0–14 years
- Mean age of children aged 5–17
- Percentage of males in the household
- Number of household members
- Percentage of children aged 1–14 who are boys
- Number of household members aged 14 and over
- Number of children aged 1–14 in the household
- Percentage of boys among children aged 5–17
- Number of children living with their mother in the household
- Number of children in the household who live with their father
- Number of children in the household whose father is alive
- Number of children in the household whose mother is alive
- Number of children aged 0–17 in the household
- Percentage of men among non-parenting adults in the household
- Percentage of women among non-parenting adults in the household
- Number of children aged 5–9 in the household
- Number of children aged 5–17 in the household
- Number of adults living in the household who are not parents of the children in the household
- Percentage of children in the household living with their father
- Number of children aged 2–4 living in the household

Integral health of the household members.

- Number of household members who have not visited a doctor in the last 3 months
- Percentage of household members with health insurance
- Number of women in the household who do not know any of the contraceptive methods mentioned in the questionnaire
- Dummy variable indicating if any woman in the household does not know any of the contraceptive methods mentioned in the questionnaire
- Whether any child aged 0–4 in the household ever had a cough in the past two weeks
- Number of women in household with prejudiced/discriminatory attitudes towards HIV
- Percentage of children aged 0–4 in the household who have had diarrhoea in the past few weeks
- Dummy variable indicating if a child has had diarrhoea in the last few weeks

Education of the household members.

- Average educational level of parents in the household
- Average years of education of adults living in the household
- Average years of education of parents living in the household
- Average education level of adults living in the household
- Average educational level of parents in the household
- Max. Educational level of household members (considering 6 categories)
- Max. Education level of household members (considering 3 categories)
- If a child aged 5–17 is overage, how many years of overage has?

Employment.

- Percentage of household members aged 14 and over having worked at least 1 h in the past week
- Number of household members aged 14 and over who worked at least 1 h during the previous week

Socioeconomic indicator.

- Wealth Index Score
- Percentage of household members receiving transfers or allowances

Housing infrastructure.

- Number of computers or tablets
- Number of rooms used for sleeping
- Dummy variable indicating if there is access to the internet at home
- Any household member owns motorcycle or scooter

Education and stimulation of young children.

- Caregiver responding that he/she thinks a child should be physically punished to properly raise or educate him/her
- Average number of children's books or picture books (for children aged 5–17)

- Average number of children's books or picture books (of children aged under 5)
- Among children aged under 5 in the household minimum number of books or pictures
- Among children aged under 5 in the household maximum number of books or pictures
- Percentage of children aged 2–4 with whom no one aged 15 or over did the following activity in the last 3 days read a book or looked at pictures in a book with him/her
- Percentage of children under 5 playing with household objects/objects found outside the household
- Dummy variable indicating if for a child aged 2–4, no one aged 15 or over did the following activity with the child in the last 3 days read books or looked at pictures in a book to (name)

Variable selection lasso-logit

In the following, we present the 55 variables that make the greatest contribution to the random forest model, grouped according to the categories: household composition, health, education, work, welfare and lifestyle.

Demographic composition of the household.

- Dummy variable indicating if for any children in the household their mother is not present
- Average age among non-parent adults in the household
- Dummy variable indicating if the head of household is the father/mother of all children aged 5 to 17 years in the household
- Percentage of children in the household living with their father
- Number of children in the household living with their mother
- Dummy variable indicating indigenous descent or affiliation of the household head
- Number of females in the household who are in the age range 15 to 19
- Average age in the household
- Number of questionnaires completed for 0–4 year olds
- Dummy variable indicating if for any children in the household their father is not alive
- Number of children 5–9 years old in the household
- Number of children aged 0–17 years in household
- Percentage of boys among children 5–17 years old
- Number of females in the household who are in the age range 20–24 years old
- Number of children in the household whose mother is alive
- Dummy variable indicating if head of household is p/mother of all children 0–17 years old in the household
- Dummy variable indicating if for any child in the household his/her father and mother are not present
- Number of females in the household who are mothers
- Percentage of boys among children 0–17 years of age
- Dummy variable indicating if any female in the household is a migrant (her country of birth is not Argentina)
- Dummy variable indicating if there is any mother in the household who is currently married or in a union
- Percentage of males in the household
- Dummy variable indicating if for any children in the household their father or mother is absent
- Dummy variable indicating if any child aged 1–14 years has a different father or mother than the others
- Dummy variable indicating if any member of the household is domestic worker
- Dummy variable indicating if any woman in the household is in the age range 45–49 years old
- Age of the household head
- Dummy variable indicating if there is any mother in the household who is previously married or in a unión
- Dummy variable indicating if the head of household is male, 0 otherwise
- Number of children 3 to 4 years old in the household
- Dummy variable indicating if if any female in the household is under 18 and married/unmarried/living with a partner, 0 otherwise
- Dummy variable indicating if there are adults who are neither parents of the household's children nor grown-up children living in the household
- If for all the children in the household their mother is alive
- There is at least 1 adult aged 65 or older
- Dummy variable indicating if any child aged 5 to 17 has a different father or mother than the rest of the children in the household
- Argentina's region Patagonia
- Language of the head of household: french

Integral health of the household members.

- Dummy variable indicating if any woman in the household has had a non-spontaneous abortion, 0 otherwise
- Dummy variable indicating if the child has difficulty hearing
- Dummy variable indicating if any woman in the household has NOT heard of HIV, 0 otherwise
- Percentage of children 2–4 years old who have functional difficulties
- Dummy variable indicating if any household member did not attend a medical consultation in the last 3 months and felt unwell in the last three months
- Proportion of household members with health coverage
- Dummy variable indicating if the child has difficulty hearing sounds like other people's voices or music
- Dummy variable indicating if any household member did not attend a medical consultation in the last 3 months
- Number of women in the household who do not know any of the contraceptive methods in the questionnaire

- Dummy variable indicating if any child aged 0–4 years in the household has had a cough at any time in the last two weeks
- Dummy variable indicating if any household member with health coverage
- Dummy variable indicating if any woman in the household had her mother's health checked in a hospital, health centre or other public place, 0 if not (base category: health check in private place)
- Number of women in the household who have felt discriminated against because of: Disability
- Dummy variable indicating if any woman in the household gave birth in a hospital, health centre or other public place, 0 if not (base category: delivery in a private place)
- Dummy variable indicating if any woman in the household was abused as a child/adolescent, 0 if not
- Dummy variable indicating if any adult other than the mother or father has taken the child to a health check-up without being sick, 0 if not
- Dummy variable indicating if you did not take your child for a health check-up because you consider it unnecessary
- Number of women in the household who have felt discriminated against because of: Sexual Orientation
- Dummy variable indicating if any member of the household is a Certified Disability Beneficiary
- Without the use of equipment or assistance the child has difficulty in walking
- Number of women in the household whose health was checked in a public hospital or health centre
- Number of women whose babies were checked at home
- Number of women in the household who had a non-spontaneous miscarriage
- Someone in the household has a Certificate of Disability
- Dummy variable indicating if a parent has taken child for health check-up without being sick
- Number of women in the household whose health was not checked after leaving the health facility after giving birth
- Dummy variable indicating if any child has any difficulties/disabilities
- Dummy variable indicating if any woman in the household had her baby's health checked in a hospital, health centre or other public place, 0 if not (base category: health check in private place)
- Dummy variable indicating if she did not take her child to a health check-up because of a money problem.
- Number of women in the household whose babies were not checked when they left the health centre
- Dummy variable indicating if any child has any visual, hearing, or walking difficulties
- Dummy variable indicating if any woman in the household did not have any prenatal check-up, 0 if she did.
- Number of women whose babies were checked in a public hospital or health centre
- Dummy variable indicating if any woman in the household uses modern methods of contraception
- Dummy variable indicating if any woman in the household had her baby's health checked at home, 0 if not (base category: health check-up in private place)

Education of the household members.

- Max. Educational level of household members (considering 3 categories)
- Dummy variable indicating if any member of the household never attended school

Employment.

- Dummy variable indicating if any household member aged 14 and overlooked for work in the last 30 days
- Number of household members over 14 years of age who worked at least 1 h in the previous week

Socioeconomic indicator.

- Dummy variable indicating if someone in the household receives Hacemos futuro (ex Argentina trabaja and Ellas Hacen) / Complementary Social Wages
- Dummy variable indicating if someone in the household receives Unemployment Insurance
- Dummy variable indicating if someone in the household receives a municipal or provincial cash transfer programme.
- Dummy variable indicating if someone in the household receives Non-Contributory Pension for Mothers of 7 children
- Household income 9th percentile
- Household income 8th percentile
- Household income 4th quartiles
- Household income 2nd quartiles
- Dummy variable indicating if all members share budget

Housing infrastructure.

- Dummy variable indicating if main source of water used for other purposes (if bottled water is used for drinking) is not piping outside the dwelling or connected to a neighbour, public tap, well with pipe, unprotected well, rain water, tanker truck, surface water, bottled water or bag of water.
- Dummy variable indicating if households using this toilet facility
- Dummy variable indicating if the place where contents were emptied is buried in a covered well
- Dummy variable indicating if in the block of the dwelling there is: waste collection
- Dummy variable indicating if the type of energy source for heating: wood/wood
- Dummy variable indicating if the type of heating in the household: Cookstove made
- Dummy variable indicating if the pit latrine or septic tank has been emptied: don't know when
- Dummy variable indicating if the type of energy source for heating: charcoal
- Dummy variable indicating if the main floor material: cement

- Dummy variable indicating if main source of water used for other purposes (if bottled water is used for drinking): piped outside the dwelling, but inside the grounds
- Dummy variable indicating if the water was treated to make it safer to drink
- Dummy variable indicating if the type of dwelling: Ranch
- Dummy variable indicating if latrine not known where the contents were emptied
- Dummy variable indicating if the main source of water used for other purposes (if bottled water is used for drinking): dug well-protected well
- Dummy variable indicating if the main material of walls: plywood
- Dummy variable indicating if the location of the sanitary installation is outside the dwelling or outside the lot
- Dummy variable indicating if the type of lighting in the household: hand lamp, lamp, or lamp
- Dummy variable indicating if a child was previously receiving AUH (cash transfer program), he/she stopped receiving it, never received it, and did not ask for it.
- Dummy variable indicating if the type of dwelling: Tenement
- Dummy variable indicating if the main wall material: adobe wall covered
- Dummy variable indicating if the main source of water for drinking: Water bag
- Dummy variable indicating if the main source of water used for other purposes (if bottled water is used for drinking): tanker truck
- Dummy variable indicating if the type of heating in the home: Manufactured space heater
- How many computers or tablets
- Dummy variable indicating if the type of cooker primarily used for cooking: No cooking in the home
- Dummy variable indicating if the main roofing material: membrane/asphaltic roofing
- Dummy variable indicating if the main roof material: board plate/cardboard
- Dummy variable indicating if the main roofing material is not cardboard sheet, cardboard, membrane, asphalt roofing, thatch, mud board, wood plank, fibre cement sheet, tile, slab, ceramic tile, or metal sheet.
- Dummy variable indicating if the type of home heating is not space heater or manufactured or traditional cooker, open fire.
- Dummy variable indicating if the type of cooker mainly used for cooking: mains gas cooker
- Dummy variable indicating if the main floor material: ceramic tiles
- Dummy variable indicating if the location of sanitary facilities: inside the dwelling itself
- Dummy variable indicating if the main source of water used for other purposes (if bottled water is used for drinking): unprotected well
- Dummy variable indicating if the type of energy source for heating: piped natural gas
- Dummy variable indicating if the type of cooker mainly used for cooking: liquid fuel cooker
- Dummy variable indicating if the household has: TV service
- Dummy variable indicating if the type of dwelling: Unbuilt room for habitation
- Dummy variable indicating if the household has: Free-standing freezer
- Dummy variable indicating if the in the block of the dwelling there is: mains gas
- Dummy variable indicating if the water treatment: add bleach / chlorine
- Dummy variable indicating if the ownership of the dwelling: lent for job
- Dummy variable indicating if the main material of walls: cane/wood/straw
- Dummy variable indicating if the main reason for not having enough water is neither that there is no mains water available, nor that water is too expensive, nor that the source is not accessible.
- Dummy variable indicating if in the block of the dwelling there is a sewage system.
- Dummy variable indicating if the main material of walls: stone
- Dummy variable indicating if the main material of walls is reused wood
- Dummy variable indicating if the household has: Television
- Dummy variable indicating if the main reason for not enough water: Source not accessible
- Dummy variable indicating if the household located on or near: A permanent rubbish dump within three blocks
- Dummy variable indicating if the type of energy source for heating: processed biomass (pellets) or wood chips
- Dummy variable indicating if the household has a wardrobe
- Dummy variable indicating if the type of cooker primarily used for cooking: solid fuel manufactured cooker
- Dummy variable indicating if the type of lighting in the home: none
- Dummy variable indicating if the type of lighting in the home is wood
- Dummy variable indicating if the type of dwelling: Cottage
- Dummy variable indicating if the type of energy source used for cooking is wood
- Dummy variable indicating if in the block of the dwelling there is: Street with sewage drainage
- Dummy variable indicating if in the household there is a mobile phone is Smartphone
- Dummy variable indicating if the type of lighting in the household: kerosen lamp
- Dummy variable indicating if the main material of the roof: fibrocement plate or plastic
- Dummy variable indicating if the main wall material is neither logs, straw, waste material, adobe, plywood, cardboard, reused wood, cement, stone, bricks, concrete block or clad adobe.

Education and stimulation of young children.

- Average number of children's books or pictures (of children under 5 years old)
- Percentage of children aged 24–59 months in the household with whom someone has conducted four or more activities to promote early stimulation and responsive care in the last 3 days
- Among children aged under 5 in the household maximum number of books or pictures
- Dummy variable indicating if any child 0–4 years old has home-made toys
- Among children aged under 5 in the household minimum number of books or pictures

- Dummy variable indicating if a child has difficulty being understood by people outside the household
- Percentage of children aged 2 to 4 years old with whom, in the past 3 days, no one older than 15 years old did the following activity with the child: Told the child stories
- Dummy variable indicating if any child under 5 years of age was supervised by another child under 10 years of age for more than one hour at least once in the last week
- Percentage of children aged 24–59 months in the household with whom the mother has done four or more activities to promote early stimulation and responsive caregiving in the last 3 days
- Dummy variable indicating if a child in the household compared to children of the same age have difficulty learning things
- Dummy variable indicating if any woman in the household in the last 12 months has felt discriminated against for any reason other than educational level, religion, age, income level, sexual orientation, or ethnicity.
- Dummy variable indicating if the child has a free lunch or free snack in a school canteen
- Percentage of children aged 2 to 4 years old with whom, in the past 3 days, no one older than 15 years old did the following activity with the child: sang songs to (name), sang with (name), including lullabies
- Dummy variable indicating if there is a child aged 24–59 months in the household with whom the mother has done four or more activities to promote early stimulation and responsive care in the past 3 days
- Availability of children's books or picture books (for children 5–17 years of age)
- Number of women in the household who have felt discriminated against because of: Religion or belief
- Percentage of children under 5 who own three or more children's books
- Dummy variable indicating if any woman in the household in the last 12 months has felt discriminated against for: any other reason
- Percentage of children aged 3–4 years who are developmentally appropriate in the indicated domains
- Dummy variable indicating if the main reasons why children under 5 did not attend school: there were no places in the nearest kindergarten, kindergarten or day-care centre in the neighbourhood.
- Number of children aged 5 to 17 in the household who never attended school
- Dummy variable indicating if in the last 12 months did not go to school because their teacher was absent
- Dummy variable indicating if in the last 12 months the school was closed for any reason other than natural disaster or teacher's strike.
- Dummy variable indicating if any child aged 5 to 17 dropped out of school because of: problems with school (expulsion, fights, etc.)
- Dummy variable indicating if the main reasons why a child under 5 does not attend an educational establishment other than my partner does not want to, it is expensive, it is far away, it is complicated to get to, there were no places in the neighbourhood kindergarten, the timetables are incompatible with my working day, I did not know about the existence of educational establishments for such young children.
- Dummy variable indicating if any child from 5 to 17 dropped out of school because: they don't consider it necessary
- Dummy variable indicating if any child from 5 to 17 stopped studying because: learning difficulties
- Dummy variable indicating if the main reasons why a child under 5 does not attend an educational institution: it is far away or difficult to get there
- Number of children aged 0–17 years in the household who never attended school
- Dummy variable indicating if any child aged 5 to 17 dropped out of school because: was ill
- Number of children aged 3 to 17 in the household who dropped out of school
- Dummy variable indicating if the main reasons for children under 5 not attending an educational institution: don't need, pay someone to look after them at home
- Dummy variable indicating if any child aged 5–17 dropped out of school because of: pregnancy (own/partner)
- Dummy variable indicating if in the last 12 months attended school for a celebration or sporting event
- Number of children aged 5 to 17 in the household who never attended school
- Dummy variable indicating if any child aged 5 to 17 dropped out of school because: there are no kindergarten/primary/middle schools in the neighbourhood
- Number of children aged 5–17 in the household who dropped out of school
- Dummy variable indicating if in the last 12 months school was closed due to natural disasters
- Dummy variable indicating if any child aged 5–17 dropped out of school because: had to go out to work
- Dummy variable indicating if in the last 12 months went to the school to talk to teachers about the child's progress

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